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MILITARY SPECIFICATION

SCALES, GRAPHICAL FIRING: SITE, M58, M67, M72, AND M86

1. SCOPE

1.1 This specification covers graphical firing site scales which provide a means of rapidly calculating firing data corresponding to the ammunition being used (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–S–12937 —Scales and Tables, Graphical Firing: Scales and Tables, Graphical Site; Packaging of.
MIL–F–13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.
MIL–I–45208 —Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedure and Tables for Inspection by Attributes.
MIL–STD–109 —Quality Assurance Terms and Definitions.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F7595190 —Scale, Graphical Firing: Site, M72
F7693989 —Scale, Graphical Firing: Site, M67
F8286788 —Scale, Graphical Firing: Site, M86
F8299951 —Scale, Graphical Firing: Site, M58

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 **Fabrication.** The scales shall be manufactured in accordance with the applicable drawings listed in Section 2 and all drawings pertaining thereto.

3.3 **General specifications.** The contractor shall be responsible for adherence to and compliance with the following requirements of Specification MIL–F–13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Part identification and marking.
(d) Workmanship.

3.4 **Environmental.**

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3.4.1 *Operating temperatures.* The scales shall be operable when they are exposed to and thermally stabilized at ambient temperatures of −40° and +150° F.

3.4.2 *Storage temperatures.* The scales shall meet the requirements of 3.5 when thermally stabilized at the standard ambient temperature of between 60° and 90° F after having been exposed to and thermally stabilized at ambient temperatures of −80° and +160° F.

3.4.3 *Humidity.* The scales shall meet the humidity requirements of Specification MIL–F–13926, Procedure I. The scales shall then be thermally stabilized after return to the standard ambient temperature of between 60° and 90° F, and shall meet the requirements of 3.5.

3.5 Post-environmental. Subsequent to the temperature and humidity tests specified in 3.4.2 and 3.4.3, the scales shall be operable and shall show no color changes, distortion, warpage, breaks, stickiness, cracks, mottling, or separation of bonded materials, and the scales shall meet all requirements of this specification.

3.6 Printing and surface quality. Letters and numbers shall be in accordance with the applicable drawings. They shall be sharp, evenly colored, without skips or breaks, and shall not touch each other or any line of any scale. The printing surface shall be free from stains and other imperfections.

3.7 Scales. All scales shall be printed in black or red as indicated on pertinent drawings. The graduations shall be accurately placed so that errors will not affect the result of computations listed in 3.9 in excess of the tolerances indicated.

3.8 Graduation error— "C" and "D" scales.

3.8.1 With the first graduation of the "C" scale in coincidence with the first graduation of the "D" scale, each succeeding graduation on the "C" scale shall be in alinement with its respective graduation on the "D" scale within 0.005 inch.

3.8.2 A 2/3 ratio between graduations on the "C" and "D" scales shall be read throughout the site scale within the tolerance indicated in 3.8.1 when the first graduation line on the "C" scale is set in coincidence with the 1.5 graduation of the "D" scale.

TABLE I. *Scale, Graphical Firing, Site: M58*

ANGLE OF SITE COMPUTATIONS

VERTICAL INTERVAL (Set hairline of indicator assembly on "D" scale)	RANGE (Yards) (Indicate under hairline)	ANGLE OF SITE (Mils) (Read on "D" scale opposite Yd or M gage point)
+125 yards	3000 (on "C" scale)	+42.49±0.10 (Yd gage point)
+75	2500 (on "C" scale)	+33.46±0.10 (M gage point)
+175	3500 (on "C" scale)	+50.99±0.10 (Yd gage point)

SITE COMPUTATIONS

VERTICAL INTERVAL (Set hairline of indicator assembly on "D" scale)	SITE RANGE (Indicate under hairline)	SITE (Mils) (Read on "D" scale opposite Yd or M gage point)
+200 meters	14,000 meters (Range slide) TAG–N	+18.11±p.10 (M gage point)
+250 yards	17,500 yards (Range slide) TAG–S	+14.69±0.10 (Yd gage point)
+200 meters	10,000 meters (Range slide) TAG–N	+21.17±0.10 (M gage point)
+200 yards	10,000 yards (Range slide) TAG–N	+20.97±0.10 (Yd gage point)
+250 meters	17,000 yards (Range slide) TAG–S	+16.48±0.10 (M gage point)
—250 yards	17,000 yards (Range slide) TAG–S	—15.00±0.10 (Yd gage point)

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3.9 Accuracy. Site and angle of site readings shall be accurate within the tolerances specified in Tables I through IV for the vertical intervals and ranges listed (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

TABLE II. *Scale, Graphical Firing, Site: M67*

ANGLE OF SITE COMPUTATIONS		
VERTICAL INTERVAL (Meters) (Set hairline of indicator assembly on "D" scale)	RANGE (Meters) (Indicate under hairline)	ANGLE OF SITE (Mils) (Read on "D" Scale opposite M gage point)
—15	2000 (on "C" scale)	—7.64±0.02
+35	5400 (on "C" scale)	+6.60±0.02
+60	7000 (on "C" scale)	+8.73±0.02

SITE COMPUTATIONS		
VERTICAL INTERVAL (Meters) (Set hairline of indicator assembly on "D" scale)	SITE RANGE (Meters) (Indicate under hairline)	SITE (Mils) (Read on "D" scale opposite M gage point)
+20	1500 (Charge 1, TAG)	+14.04±0.03
—20	2500 (Charge 1, TBG)	— 9.09±0.02
+50	2000 (Charge 3, TAG)	+26.00±0.05
—60	2500 (Charge 3, TBG)	—25.23±0.05
+75	3000 (Charge 4, TAG)	+26.23±0.05
—125	5000 (Charge 4, TBG)	—28.26±0.05
+150	4000 (Charge 5, TAG)	+38.74±0.07
—130	6000 (Charge 5, TBG)	—23.72±0.05
+150	7000 (Charge 6, TAG)	+23.02±0.05
—200	9000 (Charge 6, TBG)	—25.49±0.05
+400	8000 (Charge 7, TAG)	+52.75±0.10
—300	10000 (Charge 7, TBG)	—32.52±0.06

TABLE III. *Scale, Graphical Firing, Site: M72*

ANGLE OF SITE COMPUTATIONS		
VERTICAL INTERVAL (Meters) (Set hairline of indicator assembly on "D" scale)	RANGE (Meters) (Indicate under hairline)	ANGLE OF SITE (Mils) (Read on "D" Scale opposite M gage point)
+120	3500 (on "C" scale)	+34.93±0.07
+200	9000 (on "C" scale)	+22.63±0.05
—400	15000 (on "C" scale)	—27.17±0.05

SITE COMPUTATIONS		
VERTICAL INTERVAL (Meters) (Set hairline of indicator assembly on "D" scale)	SITE RANGE (Meters) (Indicate under hairline)	SITE (Mils) (Read on "D" scale opposite M gage point)
+50	2000 (Charge 1, TAG)	+26.05±0.05
—75	4000 (Charge 1, TBG)	—22.30±0.05
+90	3500 (Charge 3, TAG)	+27.24±0.05
—150	5000 (Charge 3, TBG)	—33.90±0.07
+80	4000 (Charge 4, TAG)	+20.77±0.04
—200	7000 (Charge 4, TBG)	—33.57±0.07
+175	5000 (Charge 5, TAG)	+36.49±0.07
—220	8000 (Charge 5, TBG)	—30.79±0.06
+250	7500 (Charge 6, TAG)	+35.53±0.07
—300	9500 (Charge 6, TBG)	—34.80±0.07
+75	8000 (Charge 7, TAG)	+9.78±0.02
—200	9000 (Charge 7, TBG)	—23.31±0.05

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

4.2 Inspection provisions.

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TABLE IV. *Scale, Graphical Firing, Site: M86*

ANGLE OF SITE COMPUTATIONS

VERTICAL INTERVAL (Meters) (Set hairline of indicator assembly on "D" scale)	RANGE (Meters) (Indicate under hairline	ANGLE OF SITE (Mils) (Read on "D" Scale opposite M gage point)
+35	2000 (on "C" scale)	+17.83±0.04
+75	4000 (on "C" scale)	+19.10±0.04
—120	8000 (on "C" scale)	—15.28±0.03

SITE COMPUTATIONS

VERTICAL INTERVAL (Meters) (Set hairline of indicator assembly on "D" scale)	SITE RANGE (Meters) (Indicate under hairline)	SITE (Mils) (Read on "D" scale opposite M gage point)
+75	3000 (Charge 1, TAG)	+27.47±0.05
—90	4500 (Charge 1, TBG)	—25.88±0.05
+80	4500 (Charge 2, TAG)	+18.82±0.04
+200	7000 (Charge 2, TAG)	+35.45±0.07
—120	4000 (Charge 2, TBG)	—31.34±0.06
—150	6500 (Charge 2, TBG)	—26.45±0.05
+200	7000 (Charge 3, TAG)	+29.88±0.06
250	12500 (Charge 3, TAG)	+26.53±0.05
—150	7500 (Charge 3, TBG)	—20.95±0.04
—200	13000 (Charge 3, TBG)	—20.75±0.04

Note: Definitions—TAG —Target above gun.
TBG —Target below gun.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105, and the inspection level and sampling plans specified in Table V titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL–E–12937 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE V. *Classification of Defects*

Use Inspection Level II of Table I with Sampling Plan Table II–A of Standard MIL–STD–105.

		Requirement	*Test Procedure*
Critical: None.			
Major: AQL 1.0 percent defective.			
101.	Printing and surface quality	3.6	4.5.5
102.	Defective scales	3.7	4.5.6
103.	Graduation error	3.8	4.5.7
104.	Accuracy failure	3.9	4.5.8
Minor: None.			

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" specified in Standard MIL–STD–105.

4.3 **Control tests.** Three graphical firing site scales shall be selected at random from each 50 produced, or from each month, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control Test	*Requirement*	*Test Procedure*
Fabrication	3.2	4.5.1
General Specifications	3.3	Visual, MIL–F–13926
Operating Temperatures	3.4.1	4.5.2
Storage Temperatures	3.4.2	4.5.2, 4.5.4
Humidity	3.4.3	4.5.3, 4.5.4

4.3.1 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the con-

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tractor and the resubmitted samples have been approved.

4.4 **Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment" and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2 *Equipment used for environmental testing.* All environmental testing equipment used to determine conformance of the environmental requirements for temperature and humidity shall conform to the "Test Facilities" requirements outlined in Specification MIL–F–13926.

4.5 **Test methods and procedures.**

4.5.1 *Fabrication.* The graphical firing site scales shall be subjected to a visual, tactile, and dimensional inspection for conformance to the applicable drawings and requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.5.2 *Storage and operating temperatures.* This test shall be conducted in accordance with Procedure I of Specification MIL–F–13926 except that the temperatures specified in 3.4.1 and 3.4.2 shall apply. While stabilized at each temperature extreme specified in 3.4.1 the scale shall be operable when normal thumb pressure is applied to move the slide and indicator assembly. The slide shall not move with normal thumb pressure when pressure is applied (with the thumb and forefinger of the other hand) against the outside edges of the site scale. Upon completion of the specified exposures and return to standard ambient temperature (60 to 90 degrees Fahrenheit), the site scale shall be subjected to the tests specified in 4.5.4 and shall conform to the requirements referenced therein.

4.5.3 *Humidity.* This test shall be conducted in accordance with Procedure I of the humidity test in Specification MIL–F–13926. Upon completion of this test the site scale shall be subjected to the tests specified in 4.5.4 and shall conform to the requirements referenced therein.

4.5.4 *Post-environmental.* Subsequent to the tests specified in 4.5.2 and 4.5.3 the site scale shall be subjected to a visual and tactile examination for defects or damage as specified in 3.5. The site scale shall then be subjected to the tests specified in 4.5.5 to 4.5.8 inclusive, and shall conform to the requirements referenced therein.

4.5.5 *Printing and surface quality.* This test consists of a visual examination of the face of the scale for imperfections of coloring applications, letters, numbers, surface quality, stains and other imperfections in accordance with the requirements of 3.6.

4.5.6 *Scales.* All scales shall be visually inspected for quality of printed graduations and to insure that colors are in accordance

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with pertinent drawings in compliance with the requirements of 3.7. The graduations shall then be inspected for accuracy of divisions and sub-divisions by being subjected to the tests specified in 4.5.8 and shall conform to the requirements specified.

4.5.7 *Graduation error—"C" and "D" scales.* Determine graduation error between the "C" and "D" scales by using a toolmaker's microscope. Lightly clamp the site scale on the microscope table. Move the center slide until the index of the "C" scale is aligned with the index of the "D" scale and clamp the slide and the rule together lightly to prevent movement. Each graduation on the "C" scale shall be checked for alignment with the opposite graduation on the "D" scale within the tolerance specified in 3.8.1.

4.5.7.1 Determine the ratio of 2/3 for the "C" scale with respect to the "D" scale using the set-up specified above. Align the index of the "C" scale opposite the 1.5 graduation of the "D" scale and clamp the slide and the rule together lightly to prevent movement. At this setting, a ratio of 2/3 shall be read throughout the site scale with the opposite graduations in alignment within the tolerance indicated in 3.8.2.

4.5.8 *Accuracy.* Site and angle of site inspection for accuracy shall be visually conducted with the aid of the 10X magnifier. Set the hairline of the indicator assembly to the specific vertical interval on the "D" scale indicated in the respective table of 3.9 for the graphical firing site scale. The center slide is then moved until the selected range of the site or angle of site computations is under the hairline of the indicator assembly. Apply pressure with the thumb and forefinger of one hand to hold the scales in this position and move the hairline of the indicator assembly to the specified gage point and read the site or angle of site on the "D" scale. The site and angle of site readings shall be accurate within the tolerances specified in 3.9

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Specification MIL–S–12937. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The graphical firing site scales covered by this specification are special slide rules used to determine site, angle of site, and vertical interval. Readings for this type of data are obtained in accordance with the "Instructions" located on the reverse side of the site scales. The scales may also be used to solve simple problems in multiplication and division.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.

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